



Motor Drive Circuits

Brushed DC • BLDC • Stepper • Servo

Drivers • Gate drivers • Current sensing • Control

Audience: Intermediate Engineering Students

Topology • MOSFETs • PWM • Sense • Feedback



Motor Types — Quick Survey

Which motor for which job

Brushed DC

Cheap, simple drive.
Wears out brushes.

Brushless DC

Long life, efficient.
Needs electronic commutation.

Stepper

Open-loop position.
Discrete steps, simple control.

Servo (hobby)

Pre-built + closed-loop.
PWM angle input.

AC induction

Industrial workhorse.
VFD drive.

PMSM / FOC

Like BLDC + sinusoidal
control. EVs, drones.

Motor Comparison

Pros, cons, applications

Motor	Pros	Cons	Apps
Brushed DC	Cheap, easy	Brushes wear, EMI	Toys, fans, RC cars
BLDC	Efficient, long life	Needs commutation HW	Drones, e-bikes, fans
Stepper	Open-loop precision	Low efficiency, hot	3D printers, CNC
Servo (RC)	Plug-and-play	Limited torque/angle	Hobby robotics
AC induction	Robust, simple	Speed not precise	Pumps, conveyors
PMSM (FOC)	Highest efficiency	Complex control	EVs, industrial CNC

H-Bridge Topology

The fundamental building block of DC motor drive

- ▶ 4 MOSFETs in an 'H' shape — two pairs (Q1+Q4, Q2+Q3)
- ▶ Q1+Q4 ON → forward; Q2+Q3 ON → reverse; opposite pair off
- ▶ Q1+Q3 (both high) = brake; Q2+Q4 (both low) = brake
- ▶ Q1+Q2 or Q3+Q4 = shoot-through → instant FET destruction (add dead-time!)
- ▶ PWM on the low-side OR diagonal switching controls speed
- ▶ Single ICs combine all 4 FETs: DRV8870, TB6612FNG, BTS7960
- ▶ For higher current: external FETs + gate driver (DRV8701, IR2104)

BLDC / PMSM 3-Phase Drive

Three half-bridges, 120° offset

- ▶ 3 half-bridges drive U, V, W stator phases
- ▶ Trapezoidal (Block) commutation: 6 steps, simpler
- ▶ Sinusoidal / FOC: smooth, efficient, more complex
- ▶ Rotor position needed: Hall sensors, encoder, or sensorless BEMF
- ▶ Common 3-phase driver ICs: DRV8313, DRV8323, L6234, MCT8316
- ▶ For high power: gate drivers (FAN7842, UCC27200) + discrete FETs
- ▶ FOC (Field-Oriented Control) — requires fast MCU (Cortex-M4F+) or dedicated motor controller

Stepper Motor Drive

Open-loop position, micro-stepping for smoothness

- ▶ Two phases (bipolar) or four (unipolar)
- ▶ Each step = $\sim 1.8^\circ$ (200 steps/rev) typical
- ▶ Micro-stepping: 1/16, 1/256 of step — smoothness comes from current sinusoid
- ▶ Constant-current drive — keeps torque constant across speed
- ▶ Common ICs: A4988, DRV8825, TMC2208 (silent), TMC5160 (high-end)
- ▶ Heating: stepper motors run warm — design thermal pad area accordingly
- ▶ Skipping steps if accelerated too fast — use trapezoidal acceleration
- ▶ StallGuard / sensorless homing — Trinamic's clever trick

Gate Drivers

Why MCU pins can't drive power MOSFETs directly

- ▶ MOSFET gate has $\sim 1\text{-}10\text{ nF}$ capacitance — needs amps to switch in nanoseconds
- ▶ MCU can sink/source maybe 20 mA — too slow $\rightarrow$ FET runs in linear region $\rightarrow$ catches fire
- ▶ Gate driver IC: amplifies MCU signal to drive gate fast
- ▶ Common: TC4427 (low-side), IR2110 (half-bridge), UCC21520 (isolated)
- ▶ Bootstrap capacitor on high-side gate driver — generates floating supply
- ▶ Charge pump for fully-on high-side at low V_{bus}
- ▶ Isolated gate drivers for high-voltage ($\geq 60\text{ V}$) for safety

MOSFET Selection

The right power transistor for your motor

Parameter	Rule of thumb	Why
Vds rating	$\geq 2\times$ max DC bus voltage	Voltage transients on coil switching
Rds(on)	Low $\rightarrow$ less heat	Conduction loss = $I^2 \times R_{ds(on)}$
Id continuous	$\geq 2\times$ motor peak current	Headroom + thermal margin
Qg (gate charge)	Low $\rightarrow$ fast switching	Switching loss + driver size
Package	DPAK / D2PAK / TO-220	Thermal handling
Body diode	Low Vf (Schottky behaviour)	Used during dead-time

Popular choices 2026

Logic-level FETs: BSC093N15NS5 (150 V/30 A), AOTL66912 (60 V/100 A), CSD18532KCS (60 V/100 A).

Motor Current Sensing

Necessary for control, fault detection, FOC

Low-side shunt

Shunt between FETs and GND.
Cheap, simple.

High-side shunt

Shunt between VBUS and FETs.
Needs differential amp.

In-line shunt

Shunt in series with motor phase.
Most expensive, most accurate.

Hall sensor

Open-loop or closed-loop
Hall IC (ACS712, AT1452).

Current transformer

AC only (not for DC).
Isolated, low-cost.

Integrated in driver

DRV8323 has built-in
amplifiers + Hall sense.

Control Modes

From simple PWM to FOC



Control hierarchy

Simple PWM

- ▶ Duty cycle = speed (brushed DC)
- ▶ No closed-loop — speed varies with load
- ▶ Cheap, low-CPU
- ▶ Used in fans, RC cars, toys
- ▶ Stall not detected

Closed-loop (PID)

- ▶ Encoder / Hall feedback → loop
- ▶ Speed control regardless of load
- ▶ Position control for servos
- ▶ Current loop inside speed loop
- ▶ Used everywhere serious

Protection

Stop the smoke before it starts

Overcurrent

Comparator + shunt
→ shutdown FETs.

Overvoltage

TVS / clamp resistor
on motor bus.

Undervoltage

Brown-out shutdown
protects logic.

Over-temp

NTC thermistor on heatsink
→ MCU watchdog.

Stall detect

BEMF or current drop.
Cut current after timeout.

Reverse polarity

Series MOSFET (P-ch) or diode
on bus input.

PCB Layout for Motor Drive

High-current and noise considerations

- ▶ Heavy copper traces (or polygons) on motor & bus current paths
- ▶ Bulk caps close to the H-bridge — bus inductance kills FETs
- ▶ Sense resistor placed for Kelvin sensing (separate sense traces)
- ▶ Gate-driver decoupling caps right at the IC pins
- ▶ Star-grounding sense returns separately from power returns
- ▶ Snubber RC across motor phases — kills voltage ringing
- ▶ Thermal pads with via stitching to bottom-side copper pour
- ▶ Separate digital ground island for MCU and logic from power ground

Common Motor-Drive Mistakes

What blue smoke teaches

- ▶ No dead-time → shoot-through → FET explosion in ms
- ▶ Bus cap too small → voltage spikes during PWM
- ▶ Inadequate gate-drive — FETs run linear → catch fire
- ▶ Logic-level FET driven at 3.3 V — partial enhancement, big losses
- ▶ No flyback diode on brushed motor — kickback kills driver
- ▶ Sense resistor without filter — current spikes alias to MCU ADC
- ▶ MCU brown-out during stall — FETs stuck on, motor burns
- ▶ Ground bounce → MCU resets when motor stalls

Off-the-Shelf Motor Driver Modules

Start with these — design custom only when needed

L298N

Classic 2A brushed bridge.
Low efficiency, hot.

TB6612FNG

1.2A bridge, modern.
Great for hobby robots.

DRV8825 / A4988

Stepper @ 2A.
3D printer standard.

TMC2209

Silent stepper, 2A.
For quiet 3D printers.

VESC

Open-source BLDC drive.
E-skateboards, e-bikes.

ODrive

FOC BLDC for robotics.
USB-controlled, high-end.

Key Takeaways

Key takeaways from this lecture

Pick the right motor

DC, stepper, BLDC, PMSM by job.

H-bridge fundamentals

Dead-time, gate drive, current

Gate drivers matter

MCU can't drive power FETs.

Sense + protect

Current monitor + OC shutdown.

Layout = reliability

Heavy copper, bulk caps, snubbers.

Use modules first

DRV/TMC/VESC then go custom.

Recommended Resources

Motor-drive learning paths

- ▶ Texas Instruments — 'Practical Motor Control' application notes
- ▶ STMicroelectronics — STM32 motor-control toolbox
- ▶ VESC project — open-source FOC firmware + hardware
- ▶ ODrive — robotics-grade BLDC drives
- ▶ Trinamic / Analog Devices stepper driver documentation
- ▶ Phil's Lab YouTube — motor drive videos
- ▶ Microchip — DT100114 'Motor Drive Basics'
- ▶ TI motor lab kits (TIDA-00355, TIDM-1014, etc.)

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ TI 'Practical Motor Control' app notes
- ▶ STM32 Motor Control Toolbox (free)
- ▶ VESC open-source project
- ▶ ODrive Robotics — modern FOC drive
- ▶ Trinamic stepper datasheets
- ▶ Phil's Lab YouTube — motor drive series
- ▶ Microchip DT100114 — Motor Drive Basics
- ▶ Texas Instruments motor lab kits

Thank you • Build something this week.